Robert Wienroeder

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M.Sc. Data Science; machine learning research at UC Berkeley, Lawrence Berkeley National Laboratory, and the DATEXIS and Cognitive Algorithms labs in Berlin. Previously a composer and music manager, Emmy-nominated for original music in Netflix's *The Queen's Gambit*

EDUCATION

M.Sc. Data Science, Berliner Hochschule für Technik 10/2022 – 07/2026
GPA 3.95/4.00 | Coursework: Algorithms and Data Structures, Machine Learning, Deep Learning, NLP, Recommender Systems, Data Visualization

Exchange Student, University of California, Berkeley 01/2025 – 06/2026
Coursework: Applied Machine Learning, Natural Language Processing, AI Safety

B.Mus. Jazz Performance, Universität der Künste Berlin 10/2015 – 09/2020
GPA 3.70/4.00 | Coursework: Music Production, Music Management, Music Theory, Performance (piano)

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, PyTorch, Scikit-learn, Matplotlib, Seaborn, Plotly), R, SQL

Concepts: Deep Learning, NLP, Gaussian Processes, Recommender Systems; End-to-end pipelines from preprocessing through training, evaluation, and deployment

Infrastructure: Docker, Kubernetes, Git

Languages: German (native), English (fluent), Spanish (conversational)

EXPERIENCE

Research Affiliate, Machine Learning and Environmental Data Science 09/2025 – 05/2026
Lawrence Berkeley National Laboratory
- Research on machine learning for environmental data, advised by Charuleka Varadharajan (LBNL) and Felix Biessmann (BHT); produced the GIST thesis project below and a manuscript targeted at HESS

Research Assistant, Machine Learning and Environmental Data Science 10/2025 – 12/2025
Cognitive Algorithms Lab, Berlin
- Scoped the groundwater forecasting project, sourced the Brandenburg well record from Germany's federal geoscience agency (BGR), and assembled the starting code base from predecessor studies with their authors

Research Assistant, Machine Learning and NLP for Medical Diagnosis 01/2024 – 09/2025
DATEXIS, Berlin
- Built the interactive demo for the group's sparse prototypical networks work (SPONGE / sproto) with two colleagues, linked from the model's Hugging Face page
- Published trained models to Hugging Face (DATEXIS/sproto, DATEXIS/CORe-clinical-diagnosis-prediction) and wrote the group's tutorial and onboarding repository for the internal Kubernetes training cluster

Freelance Musician, Composer and Music Manager 06/2009 – 09/2022
- Ran a solo practice: client acquisition, contracts, pricing, and delivery against deadlines
- Co-wrote and recorded a song for Netflix's *The Queen's Gambit*, Emmy-nominated in 2021

PROJECTS

Can Notes Alone Be Jazzy? - Interpretable melodic VAE for symbolic music 09/2026 – present
- Trained a melodic VAE on jazz and classical scores and transcriptions, balanced by format so it learns genre rather than data source
- Built a jazz axis in the latent space that steers melodies; tested in a blind listening pilot; accepted as a Late-Breaking Demo at ISMIR 2026

GIST - Groundwater Interpolation in Space and Time (M.Sc. thesis) 12/2025 – 05/2026
- Built a hybrid GRU–Gaussian Process model that forecasts groundwater levels at unmonitored wells, trained on ~1,040 wells in Germany

LEADERSHIP

Leléka and Still In The Woods - band leadership and production 2015 – 2021
- Co-founded and led both bands on the business side, running budgets, contracts, scheduling, logistics, label negotiations, and release and promotion across 4 albums, and on the artistic side, composing and arranging the repertoire
- Booked, routed, managed and produced over 200 concerts for two ensembles, coordinating up to 20 people per event across band, guest musicians, sound and light technicians, venue staff and photographers
- Produced the band's Berlin album release concert with the producer of *The Queen's Gambit* as a guest; the band's singer and I were then invited to write and record songs for the series
- Planned, drafted, shot and edited more than 10 music videos from concept to release

PUBLICATIONS AND TALKS

Can Notes Alone Be Jazzy? | **ISMIR 2026**, Late-Breaking Demo, Abu Dhabi (11/2026), accepted

GIST | Joint paper with the research groups at LBNL, the Cognitive Algorithms Lab (BHT) and BGR, based on my M.Sc. thesis (in preparation)

GIST | Talks: **HydroML 2026 Symposium, UT Austin** (05/2026); **Groundwater Modeling and More 2026, Princeton University** (06/2026)

HONORS AND AWARDS

DAAD Scholarship (HAW.International), German Academic Exchange Service, for studies at UC Berkeley 02/2026 – 05/2026
Emmy Nomination, original music for Netflix's *The Queen's Gambit* 2021
European Jazz Prodigy Prize, Jazzwoche Burghausen 2018